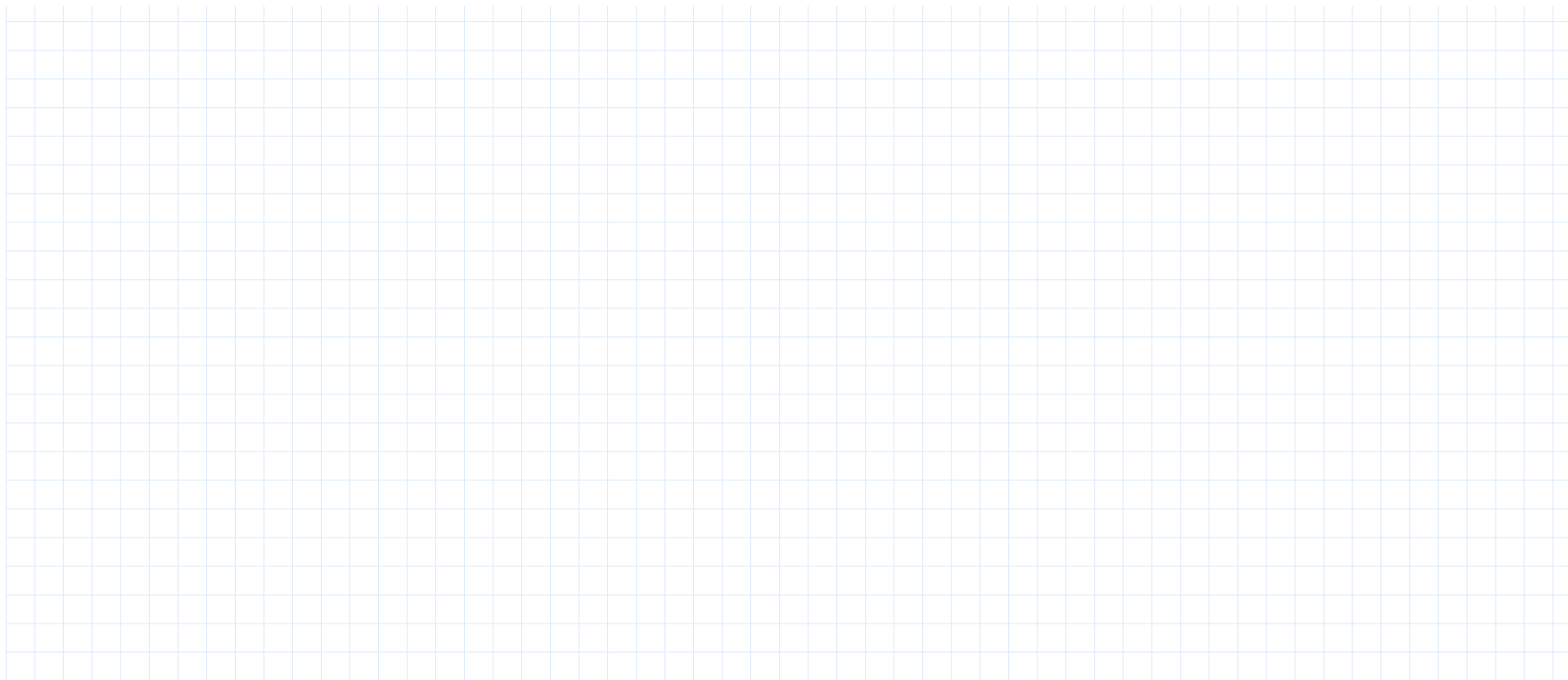


Sine and cosine repeat every 2π ; tangent repeats every π :

$$\cos(\theta + 2\pi) = \cos \theta, \quad \tan(\theta + \pi) = \tan \theta.$$

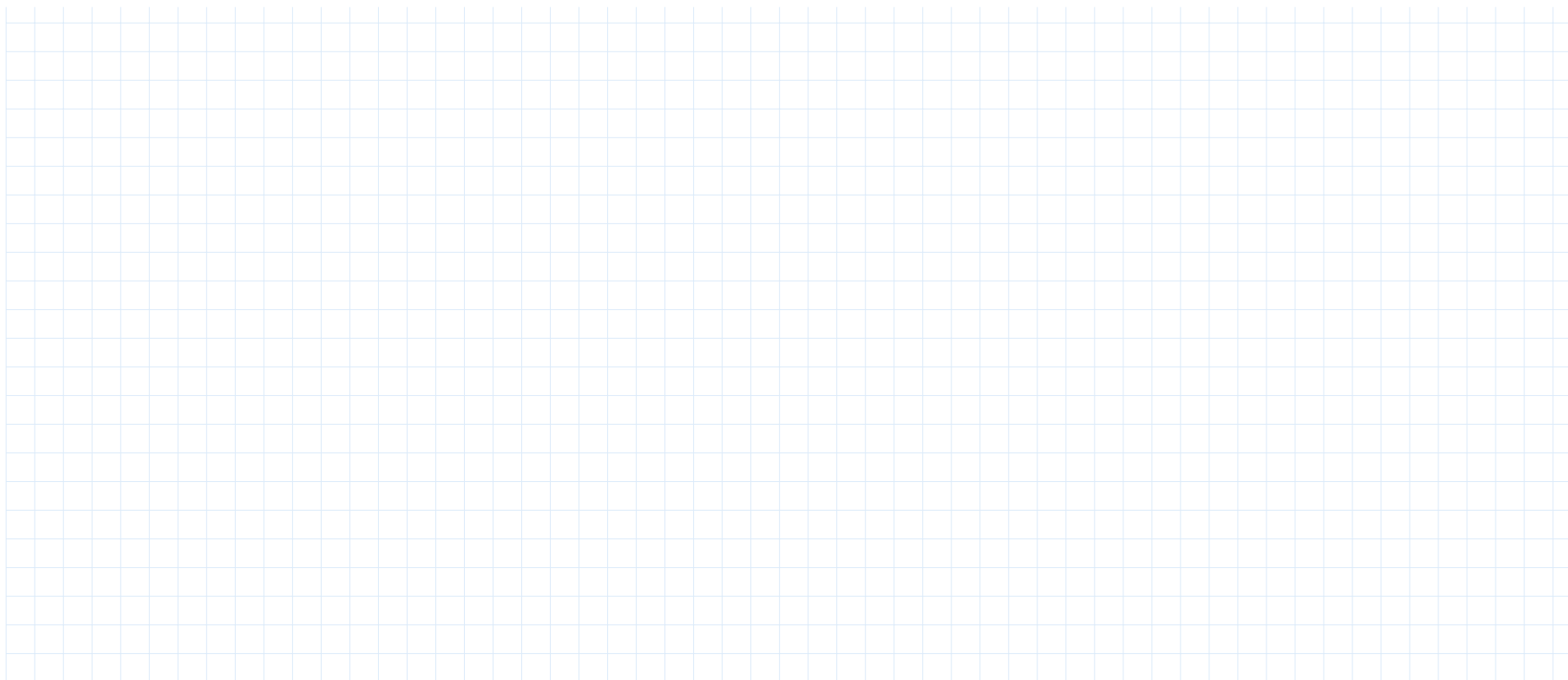
Example. Find the exact values of $\cos \frac{13\pi}{3}$ and $\tan \frac{9\pi}{4}$ by removing full periods.



Example. Suppose $\cos \theta = -\frac{5}{13}$ and $\sin \theta < 0$.

(a) Identify the quadrant. (b) Find $\sin \theta$ and $\tan \theta$.

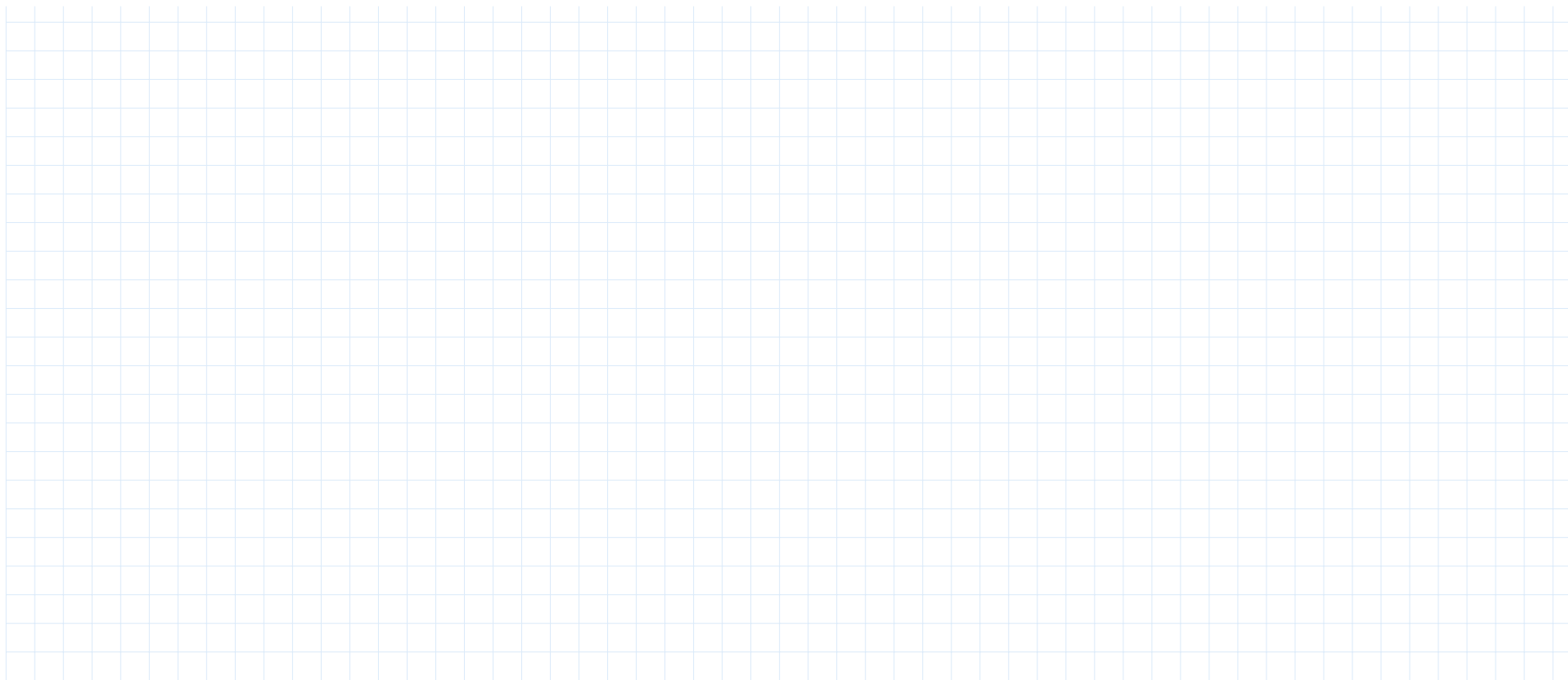
Use $\sin^2 \theta + \cos^2 \theta = 1$ and explain which sign to choose.



$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \csc \theta = \frac{1}{\sin \theta}.$$

Example. Simplify each expression. State the restriction inherited from the original expression.

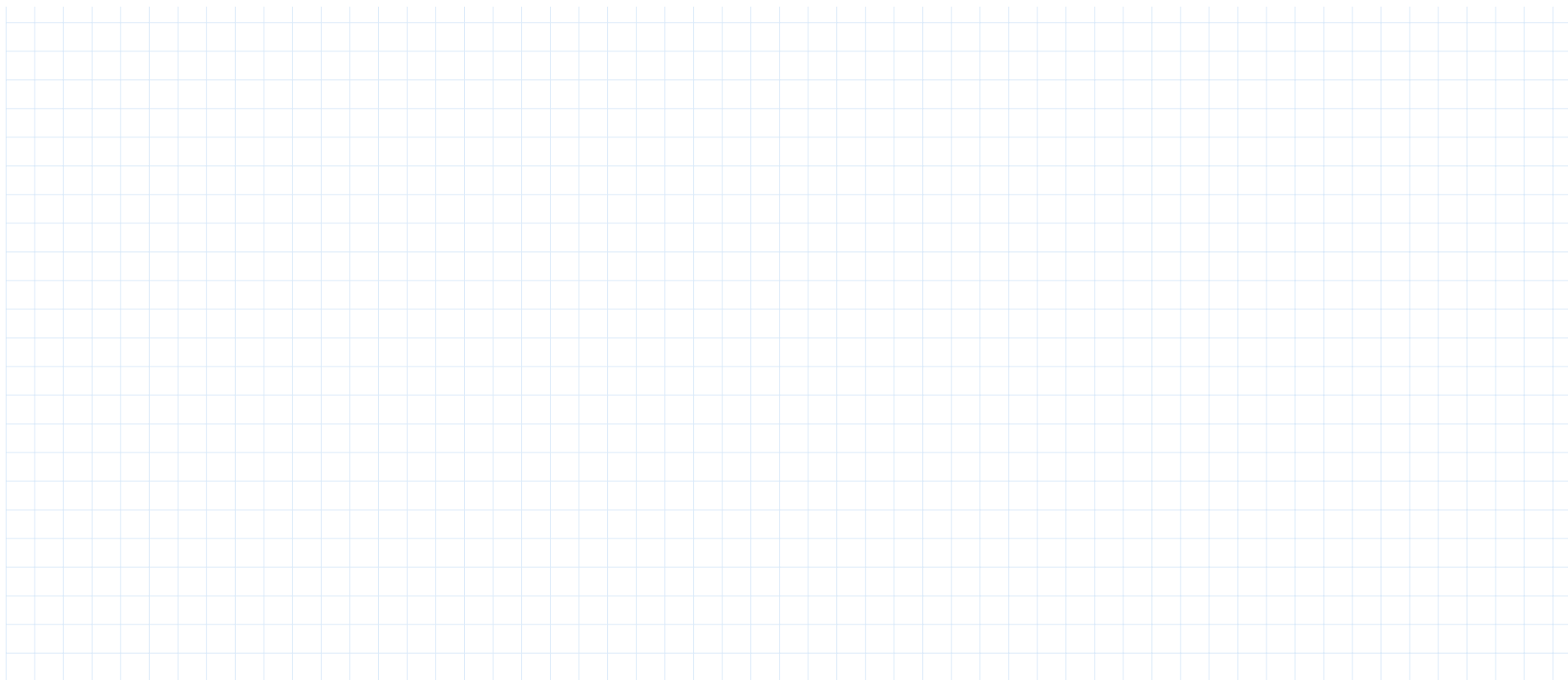
(a) $\cos \theta \csc \theta$ (b) $\tan \theta \cos \theta$.



The three forms are

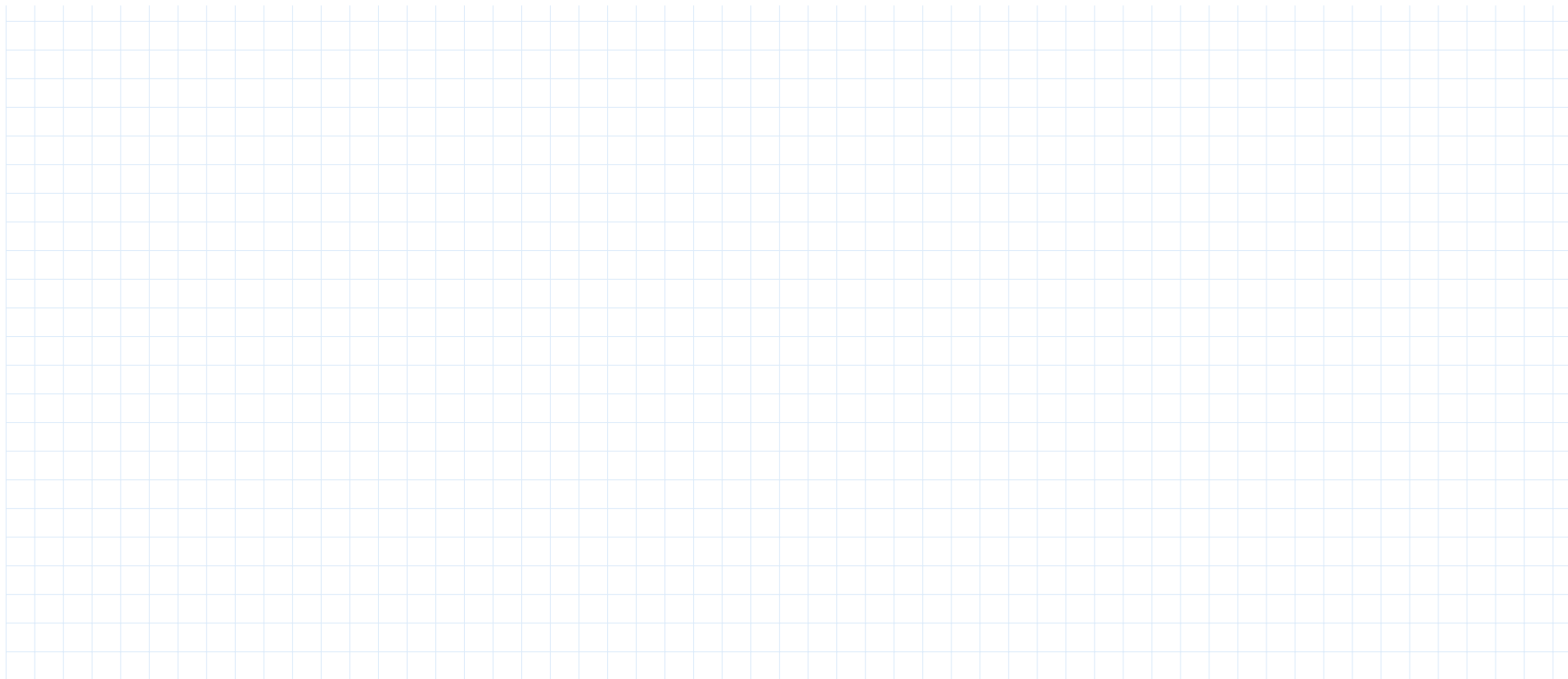
$$\sin^2 \theta + \cos^2 \theta = 1, \quad 1 + \tan^2 \theta = \sec^2 \theta, \quad 1 + \cot^2 \theta = \csc^2 \theta.$$

Example. Simplify $\sec^2 \theta - \tan^2 \theta$ and $\frac{1 - \cos^2 \theta}{\sin \theta}$. Include the original restrictions.



Recovering all six values

Example. Given $\cot \theta = -2$ and $\sin \theta > 0$, identify the quadrant and find the exact values of the other five trigonometric functions.

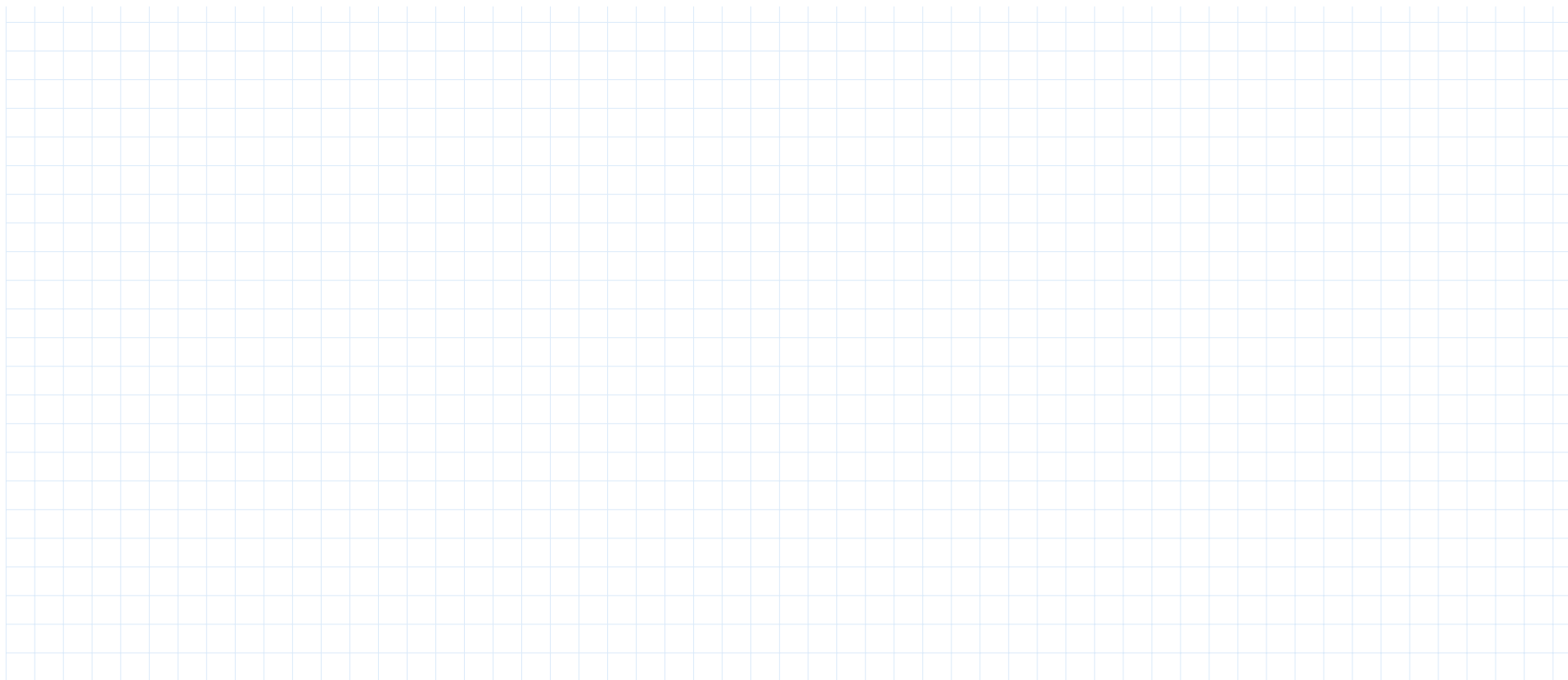


Even and odd trigonometric functions

Cosine and secant are even. Sine, tangent, cosecant, and cotangent are odd.

Example. Use these symmetries and periodicity to find exact values:

$$(a) \sin\left(-\frac{5\pi}{6}\right) \quad (b) \cos\left(-\frac{7\pi}{3}\right) \quad (c) \cot\left(-\frac{\pi}{4}\right).$$



Domains and ranges from the unit circle

A quotient is undefined when its denominator is zero. On the unit circle, both coordinates lie in $[-1, 1]$.

Example. Use these facts to state the domain and range of each function:

$$\sin \theta, \quad \cos \theta, \quad \tan \theta, \quad \cot \theta, \quad \sec \theta, \quad \csc \theta.$$

For the excluded angles, write a pattern using an integer k .

